

Tutorial Quiz 9 Rubric

Q1] Using Green's Theorem (circulation form):

$$\oint_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dA$$

Here $M = 3y$, $N = 2x$, so:

$$\frac{\partial N}{\partial x} = 2; \quad \frac{\partial M}{\partial y} = 3$$

$$\therefore \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = 2 - 3 = -1$$

$$\therefore \oint_C (3y dx + 2x dy) = \iint_R (-1) dA$$

$$= \int_0^\pi \int_0^{\sin x} (-1) dy dx$$

$$= - \int_0^\pi \sin x \cdot dx$$

$$= - [-\cos x]_0^\pi$$

$$= -2$$

- (1)

Q2]

The surface is

$$z = g(x, y) = 4 - y^2 \quad \text{with} \quad 0 \leq x \leq 1 \quad \& \quad -2 \leq y \leq 2$$

The upward normal vector is:

$$\vec{n} = \left\langle -\frac{\partial z}{\partial x}, -\frac{\partial z}{\partial y}, 1 \right\rangle$$

$$\frac{\partial z}{\partial x} = 0; \quad \frac{\partial z}{\partial y} = -2y$$

$$\therefore \vec{n} = \langle 0, 2y, 1 \rangle$$

and $\vec{F}$ on the surface z is:

$$\vec{F} = (4-y^2)^2 \hat{i} + x\hat{j} - 3(4-y^2)\hat{k}$$

The flux integral, flux = $\iint_R \vec{F} \cdot \vec{n} \, ds$ - (1)
correct
formula

$$\begin{aligned} \vec{F} \cdot \vec{n} &= (4-y^2)^2 (0) + x(2y) - 3(4-y^2)(1) \\ &= 2xy - 12 + 3y^2 \end{aligned}$$

$$\therefore \text{flux} = \int_0^1 \int_{-2}^2 (2xy - 12 + 3y^2) \, dy \, dx$$

$$= \int_0^1 -32 \, dx$$

$$= -32$$

- (1)